

Assignment

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Famous people in the world have contributed a lot to the change of the world, their words can make others change the thinking and lifestyle of many other people.

In this Exercise, you will learn about how to collect and mining on this dataset.

I. Data Collection (20 points)

Data about quotes of famous people in the world is available at the link: <http://quotes.toscrape.com/>, this website has the following interface:

The screenshot shows the 'Quotes to Scrape' website. The browser address bar displays 'Không bảo mật | quotes.toscrape.com'. The website header includes the title 'Quotes to Scrape' and a 'Login' link. The main content area features three quote cards. Each card contains a quote, the author's name with a link to their profile, and a list of tags. The first quote is by Albert Einstein with tags: change, deep-thoughts, thinking, world. The second quote is by J.K. Rowling with tags: abilities, choices. The third quote is also by Albert Einstein with tags: inspirational, life, live, miracle, miracles. On the right side, there is a 'Top Ten tags' section with a vertical list of tags: love, inspirational, life, humor, books, reading, friendship, trends, fun, and more.

Quotes to Scrape

Login

"The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking."
by [Albert Einstein](#) (about)
Tags: [change](#) [deep-thoughts](#) [thinking](#) [world](#)

"It is our choices, Harry, that show what we truly are, far more than our abilities."
by [J.K. Rowling](#) (about)
Tags: [abilities](#) [choices](#)

"There are only two ways to live your life. One is as though nothing is a miracle. The other is as though everything is a miracle."
by [Albert Einstein](#) (about)
Tags: [inspirational](#) [life](#) [live](#) [miracle](#) [miracles](#)

Top Ten tags

- love
- inspirational
- life
- humor
- books
- reading
- friendship
- trends
- fun
- more

1.1 - Please write code to crawl data from the above website, save the results to a corresponding file (kq.txt) and briefly describe the structure of the above website? **(5 points)**

1.2 - With the data you just crawled, please perform the following requests:

a. Read all the html tags (div) with the class "quote" and save it in the variable 'result', display the value of the variable 'result' on the screen? **(2 points)**

b. Find in the variable 'result', the data containing the label "small" with the class "author" and print the results on the screen? **(2 points)**

c. Write the function authorLink() to get the content of each author. For each author, print the content on the screen **(10 points)**

- Author's name
- Author's link
- Date of birth
- And the author's famous quote

d. Save the result in question c into the corresponding Quote.csv file, with each author being 1 row of data. You are required to collect at least 40 famous quotes from the above website automatically according to the code of the above ideas? **(1 point)**

II. Data Mining (60 points)

your above crawled data contains:

Field	Data Type	Description
Author	Text	Author's name
Link	Text	Author's link
DateofBirth	Date	Date of birth
Quote	Text	Author's famous quote

Table 1: Description of the dataset Quote.csv

Please write code to perform the following requirements.

2.1. Data Imputation (5 points):

- Some values of the Date of Birth field are not available, please suggest how to fill them in?
- Please add the Age field and suggest how to fill in the age of the authors?

2.2. Data Exploration (15 points):

You need to explore the data to display some statistical and analytical information of the given dataset. For example:

- Statistics about the authors and famous quotes in the dataset,
- Statistics about the year of birth and age of the authors,
- Statistics about famous quotes such as: longest, shortest, number of words, ...
- Statistics about the words used in the quotes,
- Analyze, visualize the relationship between the authors and famous quotes,
- Analyze, visualize the relationship between the authors with each other, ...

The above are just some suggestions, you can suggest more analysis and statistics.

2.3. Feature Extraction (10 points):

Suggest a way to extract features from the given dataset, provide reasons and explain your approach.

2.4. Data Mining (30 points):

You are asked to classify the statements by celebrity names and calculate the similarity of styles between authors according to the following 2 requirements:

- Predict the name of the celebrity according to the statement based on the features you extracted above and evaluate on the given dataset with Train/Test ratio and appropriate metrics? **(15 points)**
- Suggest a way to calculate the similarity of styles between authors and find the authors with the most similar styles? **(15 points)**

III. Report (20 points)

Your report is presented in each section above, using English language. The report must have a clear structure for each requirement above, with part I- you just need to provide the implementation (code), for question 1.1, point 2 you draw a structure diagram); and Part 2- each requirement includes: introduction, approach, evaluation, discussion, etc. to complete the above requirements effectively.

IV. Submission

Please submit your assignment on LMS before 11:00 PM on November 28th, 2024 (Source code + Data + Report according to the instructions in section III).
Slot 13 marking on November 29th, 2024.

